

Oral Communication Abstract 3

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Visible-Light Reductive Cyclization of Unactivated Alkyl Chlorides

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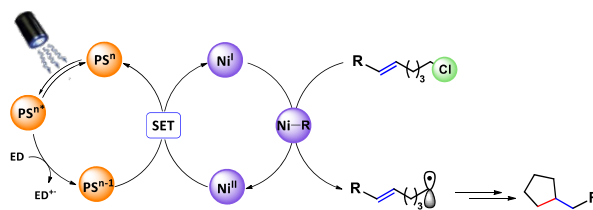
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Recent visible light metallaphotoredox methodologies have allowed the construction of a large variety of C-C and C-heteroatom bonds.¹ Most of these novel photocatalytic protocols are limited to highly active aryl and alkyl iodides and bromides as coupling partners.² Although alkyl chlorides are available, economic and bench-stable feedstocks, the chemical inertness of C_{sp3}-Cl bonds hampers their use as electrophilic partners in most of catalytic methodologies, as showcased by the narrow number of transition metal-catalyzed reactions towards this endeavor.³

In this work,⁴ we present a metallaphotoredox methodology to achieve the catalytic intramolecular reductive cyclization of unactivated alkyl chlorides with tethered alkenes. The cleavage of strong C_{sp3}-Cl bonds is mediated by a highly nucleophilic low-valent cobalt or nickel intermediate generated via visible light photoredox reduction employing a copper photosensitizer. The high basicity and multidentate nature of the ligands are key to obtain efficient metal catalysts for the functionalization of unactivated alkyl chlorides.



References

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